

Speech Synthesis

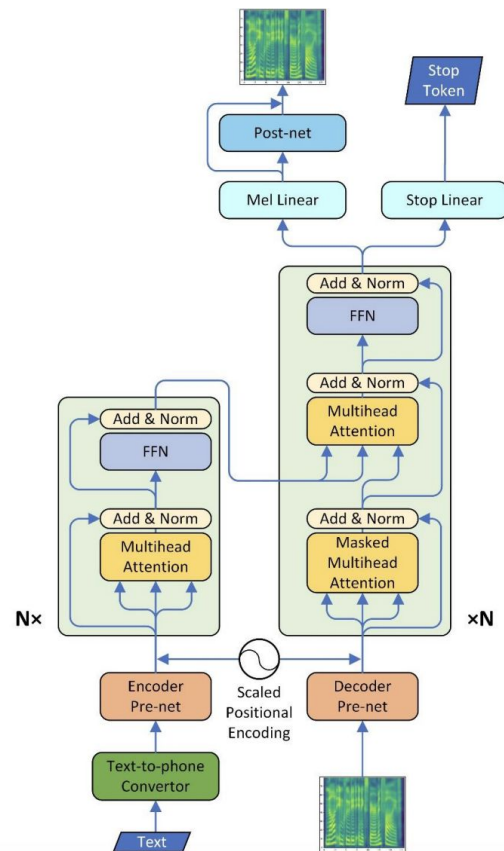
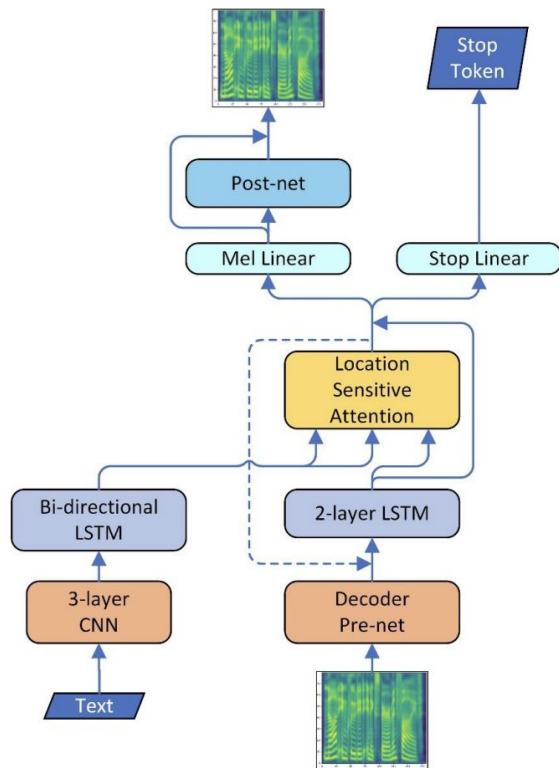
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Advanced Speech Synthesis Systems

Tacotron 2: Summary

- Converts text to acoustic feature (Mel-spectrogram)
- Encoder: To process the input text , captures linguistic features and contextual information
- Attention mechanism: Aligns the encoder and decoder states
- Decoder: To predict frame by frame Mel-spectrogram features
- Limitation: Slow inference speed due to autoregressive framework

Tacotron 2 vs Transformer TTS



Summary of WaveNet

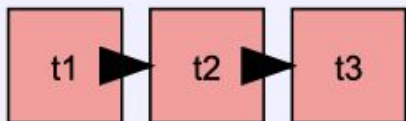
- WaveNet operates as a vocoder by transforming acoustic features (typically Mel-spectrograms) into raw waveforms.
- Wavenet Architecture consists of
 - Dilated Causal Convolutions: Kernel elements are spaced apart by a dilation rate to have longer receptive field
 - Stacked Architecture: To get deeper network
 - Residual and Skip Connections: Helps for gradient flow
 - Gated Activation Units: To control the flow of data
- Advantage: Wavenet out-performed the best TTS systems (parametric and concatenative) in Mean Opinion Scores (MOS).
- Limitation: Sequential generation process, which makes inference slow compared to newer parallel vocoder models.

Parallel WaveGan

- Yuichi, et al. proposed Parallel wavegan vocoder using GAN.
- Fast and high quality waveform generation Neural vocoder.
- It uses GAN based adversarial loss and Multiresolution STFT loss
- Supports both single speaker and multi-speakers training.
- Model size is compact with size of 11 MB.
- Inference time is real time both in CPU and GPU.

Parallel WaveGAN vs Wavenet

WaveNet Vocoder



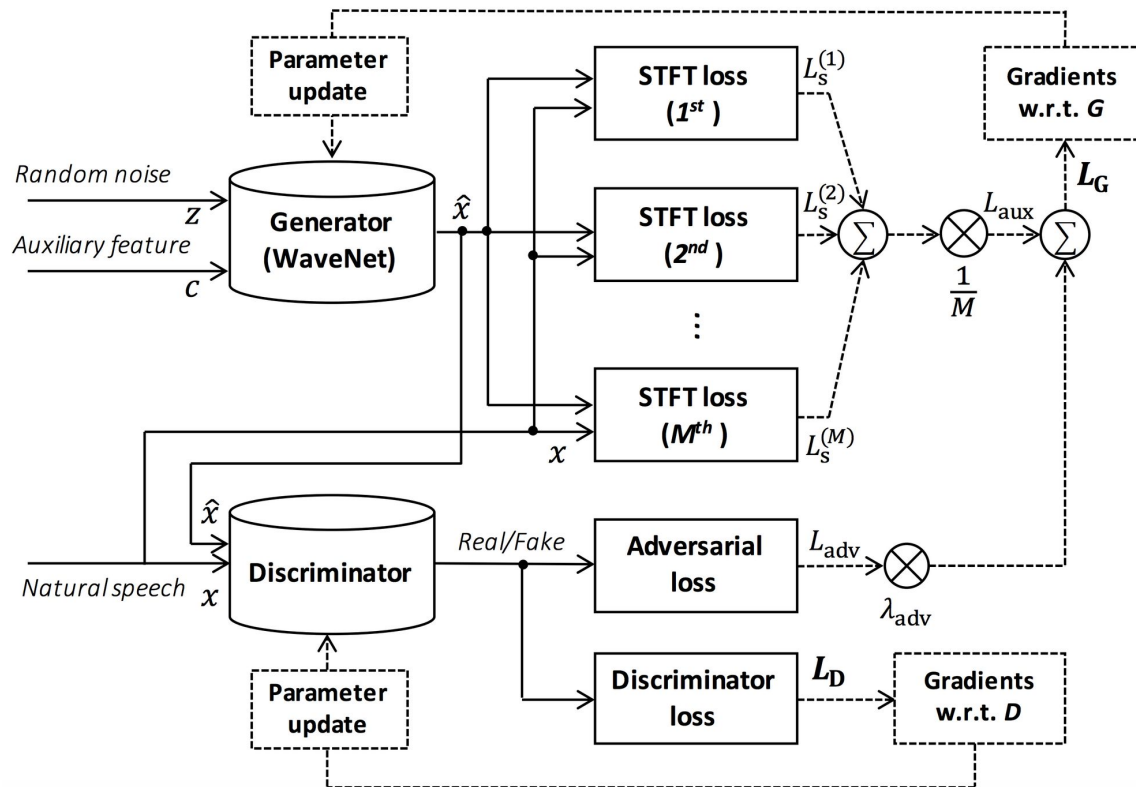
- Autoregressive generation
- High quality output
- Slow inference
- Large model size

Parallel WaveGAN



- Non-autoregressive (parallel)
- Fast inference
- Lightweight model

Parallel WaveGan Vocoder:



Parallel WaveGan for Voice cloning application

- Train the Parallel WaveGAN with large available data approximately 10 hours.
- Then fine tune for the target voice with minimum of 30 min data.
- Audio samples

Speaker	English	Hindi
Sharukh		
Modi		
Obama	 	

TTS Leaderboard

Leaderboard

Vote to help the community determine the best text-to-speech (TTS) models. The leaderboard displays models in descending order of how natural they sound (based on votes cast by the community). Important: In order to help keep results fair, the leaderboard hides results by default until the number of votes passes a threshold. Tick the [Reveal preliminary results](#) to show models without sufficient votes. Please note that preliminary results may be inaccurate.

Rank	Model	Win Rate	Votes
#1	ElevenLabs Proprietary	76.5%	18,158
#2	Kokoro v1.0	67.2%	1,410
#3	Play.ht 2.0 Proprietary	65.8%	9,722
#4	Kokoro v0.19	62.0%	3,113
#5	Play.ht 3.0 Mini Proprietary	61.8%	1,230
#6	Fish Speech v1.5	57.1%	3,244
#7	XTTSv2	56.1%	12,381
#8	StyleTTS 2	54.4%	11,681
#9	PlayDialog Proprietary	52.4%	2,325

StyleTTS2

- StyleTTS 2 is a state-of-the-art open source text-to-speech model
- Non-autoregressive architecture (generates audio in parallel)
- Combines style diffusion and adversarial training
- Supports zero-shot voice cloning and style transfer
- Trained with Single Speaker (24 hrs), multi-speaker (245 hours of audio from 1,151 speakers)
- Model parameters ~108 M, latency-> RTF: 0.0185

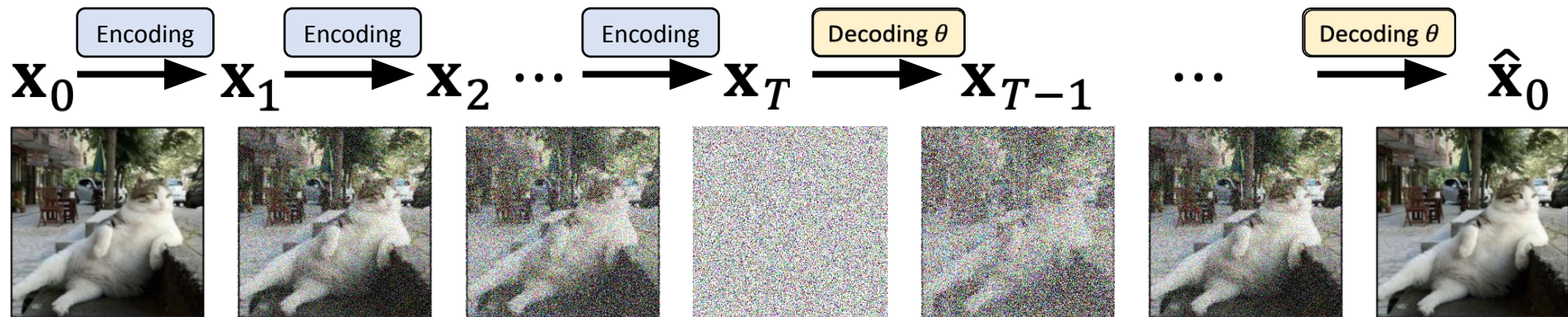
Diffusion Model: Overview

Forward Process (Noise Addition)

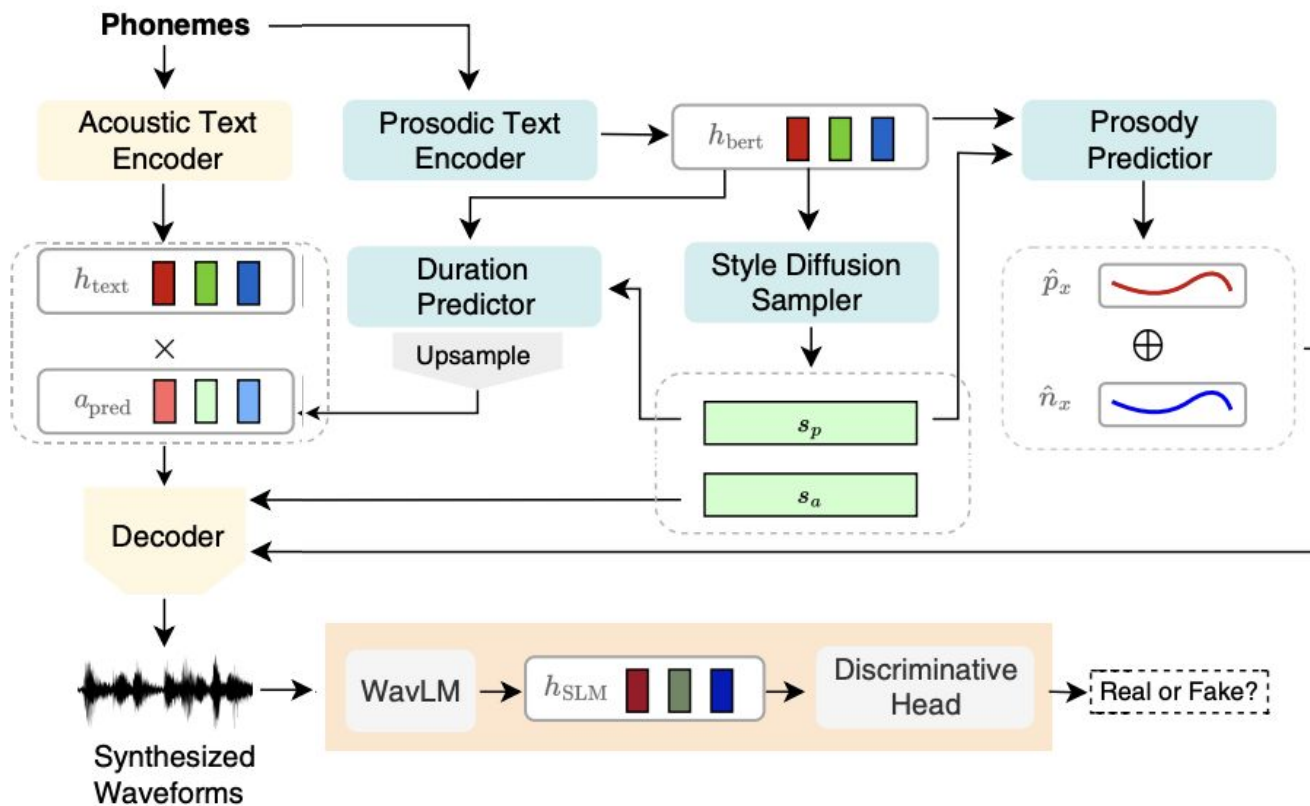
- Start with a clean mel-spectrogram (speech representation)
- Gradually add Gaussian noise over multiple steps
- Eventually transform it into pure noise

Reverse Process (Denoising)

- Start with random noise
- Gradually denoise it step by step
- Eventually recover a clean mel-spectrogram



StyleTTS2: Block diagram



StyleTTS2: Results

Table 1: Comparative mean opinion scores of naturalness and similarity for StyleTTS 2 with p-values from Wilcoxon test relative to other models. Positive scores indicate StyleTTS 2 is better.

Model	Dataset	CMOS-N (p-value)	CMOS-S (p-value)
Ground Truth	LJSpeech	+0.28 ($p = 0.021$)	—
NaturalSpeech	LJSpeech	+1.07 ($p < 10^{-6}$)	—
Ground Truth	VCTK	-0.02 ($p = 0.628$)	+0.30 ($p = 0.081$)
VITS	VCTK	+0.45 ($p = 0.009$)	+0.43 ($p = 0.032$)
Vall-E	LibriSpeech (zero-shot)	+0.67 ($p < 10^{-3}$)	-0.47 ($p < 10^{-3}$)

Table 2: Comparison of MOS with 95% confidence intervals (CI) on LJSpeech. MOS_{ID} represents MOS-N for in-distribution texts, while MOS_{OOD} is that for OOD texts.

Model	MOS_{ID} (CI)	MOS_{OOD} (CI)
Ground Truth	3.81 (± 0.09)	3.70 (± 0.11)
StyleTTS 2	3.83 ($\pm$ 0.08)	3.87 ($\pm$ 0.08)
JETS	3.57 (± 0.09)	3.21 (± 0.12)
VITS	3.34 (± 0.10)	3.21 (± 0.11)
StyleTTS + HiFiGAN	3.35 (± 0.10)	3.32 (± 0.12)

Table 3: Comparison of MOS with 95% confidence intervals (CI) on *test-clean* subset of LibriTTS for zero-shot speaker adaptation.

Model	MOS-N (CI)	MOS-S (CI)
Ground Truth	4.60 (± 0.09)	4.35 (± 0.10)
StyleTTS 2	4.15 ($\pm$ 0.11)	4.03 ($\pm$ 0.11)
YourTTS	2.35 (± 0.07)	2.42 (± 0.09)
VITS	3.69 (± 0.12)	3.54 (± 0.13)
StyleTTS + HiFi-GAN	3.91 (± 0.11)	4.01 (± 0.10)

Demo

<https://styletts2.github.io/>

Speech Synthesis – current state

- Is the standard Text-to-Speech synthesis problem solved?
 - **YES**
- Which problem is still around:
 - **Controllability**

Speech Synthesis Current Issues

- **Robustness & running cost**
 - **Out-of-distribution text:** Poor handling of unusual words, names, or technical terms.
 - **Latency concerns:** Trade-off between quality and generation speed
- **Context awareness**
 - Adaptation to user acts in dialogue and Voice characteristics can drift during long passages
 - Adaptation to the listening conditions (intelligibility)

References

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References

- ISCA Synthesis Interest Group: https://synsig.org/index.php/Main_Page (look there for the SPCC videos)
- CSTR, Simon Kings' page on speech synthesis: <http://www.speech.zone/courses/speech-synthesis/>
- CMU Festvox: <http://festvox.org/>
- HTS web page <http://hts.sp.nitech.ac.jp/> for more information on HMM based speech synthesis system (HTS)
- Google: <https://google.github.io/tacotron/>
- Espnet toolkit link: <https://github.com/espnet/espnet>

Thank You